

CURBCOUNT · REQUIREMENTS

CurbCount requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/curbcount		
Source	docs/03-requirements.md @ 141e630		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (numbered, measurable requirements with targets and concept status)
0.1	2026-09-25	Amish Chadha	Initial scaffold

CurbCount requirements

These requirements are proposed targets for concept review, awaiting Amish, and must be revised with a co-design partner before the design is frozen. Status is judged against the first-order estimates in CBC-PRC-001; "met" means met on paper only. Three requirements are **not met** by the current concept (R6 for wide sidewalks, R13 cost, and R8 and R9 if the standard FieldNode power parts are used), and two are at risk (R2, R7).

Design case: one counter on a pole 0.45 m behind the curb, sensor window 4.3 m above the road, a 3 m sidewalk, a 1.8 m bike lane and one 3.5 m traffic lane.

Table 1. Requirements.

ID	Requirement	Target	Verification	Concept status
R1	Count pedestrians, cyclists (including micromobility) and motor vehicles in the nearest lane, by direction along the street	3 classes x 2 directions	Test against manual counts	Plausible (7 to 14 frames per crossing)
R2	Count accuracy for pedestrians and cyclists	Within ± 10 % of manual counts per hour at flows up to 600 per hour per direction	Field comparison with manual counts	At risk: groups merge into one blob
R3	Count accuracy for motor vehicles in the nearest lane	Within ± 15 % of manual counts per hour	Field comparison	Unverified
R4	Privacy: no image, audio or identifier leaves the device; frames held only in RAM; sensor unable to resolve faces or plates	Ground pixel size 0.25 m or larger at the design height	Design review, firmware audit, geometry	Met by design (about 0.3 to 0.4 m)
R5	Output format	15-minute counts per class and direction in an open, documented format over LoRaWAN	Design review	Met by design (14 bytes per bin)
R6	Coverage from one pole	Full width of a 3 m sidewalk, a 2 m bike lane and the nearest lane	Geometry, field check	Not met: about 2.1 m behind the curb covered
R7	Operating range	-20 °C to +50 °C; counting works with air up to 35 °C	Thermal chamber, hot-season field test	At risk: little thermal contrast near body temperature
R8	Autonomy without sun	3 days of continuous counting	Energy budget, bench test	Met with two cells (about 4.3 days); not met with one (about 2.1 days)
R9	Energy neutral in winter	At 1.5 peak sun hours with street shading derating	Energy budget, field log	Met with 20 W panel (about 16 Wh/day for 7.2 Wh/day); not met with 6 W
R10	Ingress protection	IP65 for the enclosure and the sensor head	Spray test	By design, unverified
R11	Mass on the pole	6 kg or less	Weighing	Met (about 4.2 kg)
R12	Installation	Two trained people, 60 min, no drilling, welding or pole wiring; withstands 35 m/s gusts	Timed trial, bracket calculation	By design; wind load about 170 N to check
R13	Parts cost	\$150 or less per counter	Priced BOM	Not met: about \$271
R14	Service life	5 years outdoors with one battery change	Design review, UV-stable materials	Unverified

ID	Requirement	Target	Verification	Concept status
R15	Radio use	Within EU868 1 % duty cycle and the US915 dwell limits	Airtime calculation	Met (about 0.8 s/h against 36 s/h)

Table 2. Assumptions behind the targets.

Assumption	Value	Source or basis
Walking speed	1.0 to 1.5 m/s	Typical design range (estimate)
Cyclist speed	up to 25 km/h	Typical urban maximum (estimate)
Vehicle speed in the nearest lane	up to 50 km/h	Common urban limit (estimate)
Peak sun hours, winter design case	1.5 h	Mid-latitude winter (estimate)
Street shading and dust derating	40 %	Estimate, to be checked on real poles